

Informative dropout and trial-design choice in aggregated N-of-1 biomarker-validation trials

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Contents

1 Abstract

Background. Aggregated N-of-1 trial designs vary substantially in their robustness to informative dropout, yet the published methodological literature has documented dropout as a generic problem without adequately engaging the coupling between the trial-design family (open-label, open-label plus blinded discontinuation, traditional crossover, hybrid) and the inferential consequences of biased dropout for the biomarker-treatment interaction estimand. The unaddressed coupling is not a minor sensitivity issue: the bias and power consequences of dropout differ by an order of magnitude across design families even at a fixed overall dropout rate, and the most powerful design under no dropout is not necessarily the most powerful design under dropout.

Methods. We formalise the informative-dropout model introduced in[?] (hazard depending on cumulative symptom worsening since baseline), place it within the classical missing-data taxonomy, and then reproduce and extend Hendrickson’s four-design dropout-robustness analysis with wider parameter ranges, sensitivity to dropout-pattern misspecification at the analysis stage, and an explicit randomisation-path decomposition. The simulation employs the existing pmsimstats framework with extensions to the dropout-pattern grid, and reporting follows the[?] ADEMP framework.

Results. [Forthcoming.] Across all four design families, informative dropout that selectively retains treatment-non-responders appears to inflate the apparent biomarker-treatment interaction by underestimating the on-drug response of the responder stratum. The OL design is the most dropout-robust on power but contributes no useful biomarker-prediction information; the OL+BDC and hybrid N-of-1 designs are the most dropout-vulnerable on raw power but, paradoxically, recover power through the so-called ‘happy accident’ selection effect (biased dropout preferentially preserves the most-informative crossover blocks). The traditional crossover design is most robust on combined power-and-bias under symptom-worsening-driven dropout but most vulnerable under

35 symptom-improvement-driven dropout, where treatment-responders drop early
36 and the crossover signal is degraded.

37 **Conclusions.** Trial designers should evaluate proposed N-of-1 designs against
38 the anticipated dropout-pattern mechanism, not merely against a generic dropout
39 rate. We provide a design-by-dropout-pattern recommendation table that maps
40 three classes of dropout mechanism (balanced, worsening-driven, improvement-
41 driven) to preferred design choices.

42 2 Introduction

43 2.1 A claim about how N-of-1 trials should account for dropout

44 We begin with two coupled methodological claims about informative dropout in
45 aggregated N-of-1 trial design.

46 **The paper advances two coupled claims about how N-of-1 design selection**
47 **should account for informative dropout.**

- 48 • The four design families (OL, OL+BDC, traditional crossover, hybrid) differ
49 in dropout robustness in ways the published literature has not adequately
50 characterised.
- 51 • Hendrickson's Figure 4A demonstrates power losses exceeding 50 percent
52 in some design-by-dropout cells while remaining negligible in others.
- 53 • The dropout-by-design coupling is a first-order design choice, not a
54 marginal sensitivity issue.

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62 *We begin with two coupled methodological claims about informative dropout*
63 *in aggregated N-of-1 trial design.*

64 **The first claim is that design families differ in dropout robustness in ways**
65 **the published literature has not adequately characterised.**

- 66 • Power loss from a realistic dropout pattern can exceed 50 percent in some
67 design-by-dropout cells.
- 68 • Design rankings reverse depending on the dropout pattern.
- 69 • This is a first-order design choice, not a marginal sensitivity issue.

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77 *The first is that the four candidate trial-design families (open-label, open-label*
78 *plus blinded discontinuation, traditional crossover, and the hybrid N-of-1 de-*
79 *sign proposed by @hendrickson2020) differ in their robustness to informative*
80 *dropout in ways that the published methodological literature has not adequately*
81 *characterised. Hendrickson's Figure 4A heatmap demonstrates that the abso-*
82 *lute power loss from a realistic dropout pattern can exceed 50 percent in some*
83 *design-by-dropout cells while remaining negligible in others, and the design*
84 *rankings reverse depending on the dropout pattern. This is not a marginal*
85 *sensitivity issue but a first-order design choice that the field has not, to the*
86 *best of our knowledge, foregrounded.*

87 **The second claim is that the dropout mechanism, not merely the rate, is**
88 **the methodologically relevant object for design selection.**

- 89 • Balanced, worsening-driven, and improvement-driven mechanisms interact
90 with design families in qualitatively different ways.
- 91 • Trial designers should evaluate designs against the anticipated dropout
92 mechanism rather than a generic rate.
- 93 • The mechanism is the primary object; the rate is relatively superficial.

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101 *The second claim is that the dropout-pattern mechanism (balanced, symptom-*
102 *worsening-driven, symptom-improvement-driven) interacts with the design*
103 *family in qualitatively different ways than the overall dropout rate alone, and*
104 *that trial designers should therefore evaluate proposed N-of-1 designs against*
105 *the anticipated dropout-pattern mechanism rather than against a generic*
106 *dropout rate. That is, the mechanism is the methodologically relevant ob-*
107 *ject; the rate is relatively superficial.*

108 **The paper imports the mature missing-data taxonomy into the N-of-1**
109 **design-comparative context for the first time.**

- 110 • MCAR/MAR/MNAR taxonomy and MMRM have been worked through
111 for parallel-group RCTs but not for N-of-1 designs comparatively.
- 112 • ICH E9 (R1) formalises estimand-aware handling of intercurrent events
113 including dropout.
- 114 • The present paper undertakes the import of this machinery into the N-of-1
115 design-comparative setting.

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